Business Rule

ID	Module	Related UC	Rule Definition
BR-GEN-02	General	UC01-UC24	Users may only access functions allowed for their assigned role.
BR-ACC-03	Auth, Profiles & Master Data	UC02, UC09	Guest ID/Passport information must be collected before check-in.
BR-ACC-04	Auth, Profiles & Master Data	UC02, UC05	Personal and sensitive data may only be processed with explicit guest consent.
BR-ACC-05	Auth, Profiles & Master Data	UC05	A guest may request deletion of personal data after their stay has been completed.
BR-MST-01	Auth, Profiles & Master Data	UC03, UC04	Only authorized management roles may manage master data such as rooms, menus, tours, and staff accounts.
BR-ROOM-01	Room Reservation & Front Desk	UC06, UC07	Only rooms with Available status may be booked by guests.
BR-ROOM-02	Room Reservation & Front Desk	UC06, UC07	Check-in date must be later than the booking creation date/current date.
BR-ROOM-03	Room Reservation & Front Desk	UC06, UC07	Check-out date must be later than the check-in date.
BR-ROOM-04	Room Reservation & Front Desk	UC06, UC07	A specific room must not be booked by more than one guest for overlapping dates/times.
BR-ROOM-05	Room Reservation & Front Desk	UC07	A room booking is confirmed only after

			successful deposit payment.
BR-ROOM-06	Room Reservation & Front Desk	UC07	Room deposits are non-refundable if cancellation occurs within 48 hours before check-in.
BR-ROOM-07	Room Reservation & Front Desk	UC07	Room deposits are 100% refundable if cancellation occurs more than 48 hours before check-in.
BR-CIO-01	Room Reservation & Front Desk	UC08, UC09	Receptionists may check in only confirmed bookings.
BR-CIO-02	Room Reservation & Front Desk	UC09	A physical room number must be assigned during check-in.
BR-CIO-03	Room Reservation & Front Desk	UC09, UC10	After check-in, the physical room status must become Occupied.
BR-CIO-04	Room Reservation & Front Desk	UC09, UC10	After check-out, the physical room status must become Needs Cleaning.
BR-CIO-05	Consolidated Billing & Dashboards	UC09, UC20, UC21	A guest cannot be checked out if unpaid food orders or tour bookings are attached to the room.
BR-FOOD-01	Restaurant POS & Room Service	UC11	Only checked-in guests may place room-service food orders.
BR-FOOD-03	Restaurant POS & Room Service	UC13	Food order status must follow Pending → Preparing/Cooking → Delivered/Served.
BR-FOOD-05	Restaurant POS & Room Service	UC14, UC20	Delivered room-service food charges must be added to the guest's consolidated invoice.

BR-TABLE-01	Restaurant POS & Room Service	UC12	A specific restaurant table must not be reserved by more than one guest for overlapping dates/times.
BR-TOUR-02	Tour Booking & Feedback	UC16	Number of participants must not exceed the tour capacity.
BR-TOUR-03	Tour Booking & Feedback	UC16	Tour booking date must be before the tour/activity start date.
BR-TOUR-04	Tour Booking & Feedback	UC17	Tour Guide may only view attendee lists for tours assigned to them.
BR-REV-01	Tour Booking & Feedback	UC18	Guests may submit ratings/reviews only after completing a stay or tour.
BR-REV-03	Tour Booking & Feedback	UC18	Each completed booking or tour participation may receive at most one review from the same guest.
BR-REV-04	Tour Booking & Feedback	UC19	Only authorized management roles may moderate guest reviews.
BR-BILL-01	Consolidated Billing & Dashboards	UC20	At check-out, the system must generate a consolidated invoice.
BR-BILL-02	Consolidated Billing & Dashboards	UC20	The consolidated invoice must include room costs, unpaid food orders, and unpaid tour bookings.
BR-REP-01	Consolidated Billing & Dashboards	UC22, UC23	Revenue reports must classify revenue by Rooms, Food & Beverage, and Tours.
BR-REP-04	Consolidated Billing & Dashboards	UC22-UC24	Only authorized management roles may

			access financial dashboards and reports.

ID	Rule Definition
BR-01	A room shall not be booked by multiple guests during overlapping periods.
BR-02	A booking shall be confirmed only after successful deposit payment.
BR-03	Only rooms with status 'Available' shall be eligible for booking.
BR-04	Receptionists shall only perform check-in for confirmed bookings.
BR-05	Guests shall provide valid identification documents before check-in.
BR-06	Room status shall automatically change to 'Occupied' after check-in.
BR-07	Room status shall automatically change to 'Needs Cleaning' after check-out.
BR-08	A guest shall not be checked out while unpaid charges remain on the account.
BR-09	Only checked-in guests shall be allowed to place room service food orders.
BR-10	Food orders shall follow the workflow: Pending → Cooking → Delivered.
BR-11	Delivered food orders shall automatically be added to the guest's consolidated invoice.
BR-12	Tour bookings shall not exceed the maximum tour capacity.
BR-13	Guests shall only be allowed to submit reviews after completing their stay or tour.
BR-14	Each booking shall be allowed to submit only one review.
BR-15	The system shall generate a consolidated invoice during guest check-out.
BR-16	A consolidated invoice shall include room charges, food orders, and tour bookings.

BR-17	An invoice shall be marked as paid only after successful payment processing.
BR-18	Room deposits shall be fully refundable only if cancellation occurs more than 48 hours before check-in.
BR-19	Room deposits shall not be refunded if cancellation occurs within 48 hours before check-in.
BR-20	Revenue reports shall classify income into Room Revenue, Food & Beverage Revenue, and Tour Revenue.

ID	Business Rule	Business Rule Description	Use Case
FR01	Account Uniqueness	Each newly registered account must not have a duplicate email address or phone number with any existing account in the system.	Register Account
FR02	Password Security	User passwords, when created or changed, must be hashed securely in one direction (using Bcrypt or SHA-256) before being saved to the database. Do not store passwords in plain text.	Register Account, Change Password
FR03	ID File Specification	The identification document file uploaded to the system must be in the correct image format (.jpeg, .png) or document format (.pdf) with a	Upload CCCD/Passport

		maximum file size of no more than 5MB to ensure storage performance.	
FR04	Booking Hold Timeout	When a customer makes a reservation, the system will temporarily lock the room and hold the booking for a maximum of 15 minutes while awaiting a status update from the payment gateway. After this time, the reservation will be automatically canceled and the room will be released to "Available" status.	Book Room, Online Payment
FR05	Tour Status Lifecycle	The lifecycle state of a tour can only be updated progressively in a linear fashion: Not Started → In Progress → Completed. The system absolutely blocks reverse state updates (e.g., Completed reverting to In Progress).	Update Tour Status
FR06	Check-in Boundary	The number of customers who check in to participate in the tour in person must not exceed the total number of customers who have successfully registered and paid for that tour.	Check-in Participants, Check Number Of Guests
FR07	Table Reservation Hold	Table reservations at the restaurant are only held for a	Reserve Restaurant Table, Update Table Status

		maximum of 30 minutes from the time the customer schedules the reservation on the system. After 30 minutes, the system automatically cancels the reservation and changes the table status to "Available" to serve other customers.	
FR08	Room Charge Validation	Customers are only allowed to choose the "Debit to Room" payment method when their account has completed the check-in process at the reception desk and the Guest Folio is in an active state.	Order Food To Room, Charge Food Bill To Room
FR09	Review Permission	The system only activates the rating and scoring feature, allowing users to submit data, when and only when the Invoice ID associated with that Room or Tour has changed to the "Paid" status.	Submit Review & Rating, View Invoice
FR10	System Backup Window	The system's automatic data backup function must be scheduled at 2:00 AM daily (off-peak hours). The administrator has the right to perform manual backups at any time if system upgrades are required.	Backup Database
FR11	Database Isolation	When performing data recovery, the system is required	Restore Database

		to automatically terminate all current sessions for all Customers and Employees, and switch the entire system to "Maintenance" mode to avoid conflicts or overwriting corrupted data.	
FR12	Payment Integration	The booking process requires the inclusion of the Online Payment service. The system will only issue the official Booking ID after receiving the Transaction Hash from the partner payment gateway.	Book Room, Online Payment
FR13	Room Detail Dependency	The Room Details feature is an extension of the Room Search feature. The system only displays detailed information when the check-in/check-out date parameters have been correctly defined to calculate the accurate price at the time.	Search Room, Detail Room
FR14	Cross-Department KOT Sync	When F&B staff confirm a Process Food Order, the system is required to automatically generate a Manage KOT (Know Your Own Food Order) on the Kitchen Staff's screen within a maximum of 3 seconds to ensure real-time accuracy.	Process Food Order, Manage KOT (Receive & Confirm)

PART 2: SYSTEM-WIDE BUSINESS RULES CLASSIFICATION TABLE

This table summarizes the rules, constraints, data facts, and calculation formulas applied throughout the project's business core.

ID	Category	Rule Definition
BR-01	Constraints	The chosen check-in time or tour departure date cannot be a date in the past or before the present time.
BR-02	Constraints	When an account is marked "Blocked" by an administrator, all access and new service booking privileges for that account will be immediately suspended, and pending orders will be paused for review.
BR-03	Constraints	Guests cannot order food to their room outside of the kitchen's designated operating hours (for example, the kitchen only accepts orders from 6:00 AM to 10:00 PM daily).
BR-04	Constraints	The new password (Change Password) must not be the same as the current password and must be at least 8 characters long, including at least one uppercase letter, one lowercase letter, one number, and one special character.
BR-05	Constraints	The order status displayed to customers and F&B staff (Update Order Status) must be locked according to the kitchen's logic: Pending → Cooking → Delivered. The "Delivered" button should only light up when the system confirms that the order has been completed in the kitchen.
BR-06	Facts	A tour guide only has the right to view details and the list of participants for tours that the system or the administrator has assigned them to manage.

BR-07	Facts	All kitchen order tickets (KOTs) sent from the F&B Staff system will be sorted and processed in the kitchen according to the FIFO (First In, First Out) queuing principle to ensure fairness.
BR-08	Facts	All dishes ordered by a customer in a single order must be served to the same location (either the same room number or the same table number at the restaurant).
BR-09	Facts	Each invoice, once issued, is unique and financial information cannot be modified after it has been marked "Paid". Any subsequent modifications must be processed using a refund invoice or adjustment invoice approved by the Admin.
BR-10	Facts	The Admin's Sort Account and Search Account functions only classify and display data currently being exported from the View Account function; they do not change or delete any original records in the database.
BR-11	Computations	
BR-12	Computations	
BR-13	Computations	
BR-14	Computations	